

Supplementary File S1 - Dataset harmonization and endpoint verification

Submission-aligned audit; live GEO Series Matrix rechecked 2026-08-07

Cohort	Role	Platform	Total N	pCR	RD	Endpoint / exclusion note
GSE25055	Discovery / internal CV	GPL96	310	57	249	4 missing response labels excluded; 306 usable.
GSE25065	Same-family external	GPL96	198	42	140	16 missing response labels excluded; 182 usable.
GSE41998	Cross-platform sensitivity	GPL571	279	69	184	20 literal zeros + 6 missing excluded; 253 explicit Yes/No cases.
GSE20194	Documented, not analyzed	GPL96	278	56	222	Held out because of MDACC-lineage overlap risk.
GSE20271	Documented, not analyzed	GPL96	178	26	152	Held out because of MDACC-lineage overlap risk.

Verified GSE41998 response rule

- 1
- Live series-matrix distribution: 69 Yes, 184 No, 20 literal 0, and 6 missing among 279 samples.
- 2
- All 20 zeros remain a separate zero category in the parallel pccrcb1 field; they are not equivalent to explicit No.
- 3
- Analysis mapping: Yes -> pCR; No -> RD; literal 0 and missing -> excluded. Usable N = 253.

Interpretation boundary

GSE41998 is a cross-platform transportability sensitivity cohort. Its results do not estimate leakage and do not establish a clinical biomarker.